



PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE : MEB-I (A)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

Test Booklet Code

This Booklet contains **28** pages.

E7

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject. **Candidates are advised to read all 15 questions in each subject of Section B** before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
- Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
- No candidate, without special permission of the Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

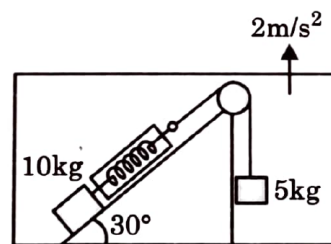
Your Target is to secure Good Rank in Pre-Medical 2024

Topic : MAGNETIC EFFECT OF CURRENT, NLM AND FRICTION, WPE [BACK UNIT] : WAVE OPTICS AND UNIT DIMENSION ERROR

SECTION-A (PHYSICS)

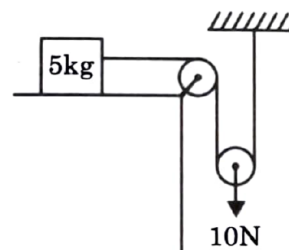
- In Young's double slit experiment, if the slit widths are in the ratio 1 : 9, then the ratio of the intensity at minima to that at maxima will be :-
 (1) 1 (2) 1/9
 (3) 1/4 (4) 1/3
- In a Young's double slit experiment, the fringe width is found to be 0.4 mm. If whole apparatus is immersed in water of refractive index $4/3$, without disturbing geometrical arrangement, new fringe width will be :-
 (1) 0.30 mm (2) 0.40 mm
 (3) 0.53 mm (4) 0.45 mm
- Light of wavelength λ is incident on a slit of width 'd'. The resulting diffraction pattern is observed on a screen at a distance D. The linear width of the principal maximum is equal to the width of the slit then 'D' equals :-
 (1) $\frac{d}{\lambda}$ (2) $\frac{2\lambda}{d}$
 (3) $\frac{d^2}{2\lambda}$ (4) $\frac{2\lambda}{d}$
- Two Nicols are oriented with their principal planes making an angle 60° . The percentage of incident unpolarized light which passes through the system is—
 (1) 50% (2) 100%
 (3) 12.5% (4) 37.5%
- Plane microwaves are incident on a long slit having a width of 5 cm. Calculate the wavelength of the microwaves if the first diffraction minimum is formed at $\theta = 30^\circ$?
 (1) 10 cm (2) 2.5 cm
 (3) 5 cm (4) None of above

- In the figure the reading of the spring balanced will be: ($g = 10\text{m/s}^2$)



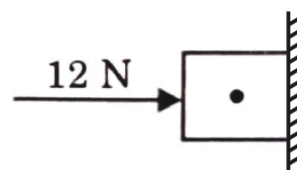
- (1) 50N (2) 40N
(3) 60N (4) 70N

- Consider the figure, pulley and strings are massless and frictionless. The horizontal surface is smooth. The acceleration of the block :



- (1) 2m/s^2 (2) 1m/s^2
(3) 0.5m/s^2 (4) 4m/s^2

- A block of weight 5 N is pushed against a vertical wall by a force 12N. The coefficient of friction between the wall and block is 0.5. The magnitude of the force exerted by the wall on the block is :



- (1) 12 N (2) 5 N
(3) 6 N (4) 13 N

- A marble block of mass 2 kg lying on ice when given a velocity of 6m/s is stopped by friction in 10s. Then the coefficient of friction is :

- (1) 0.01 (2) 0.02
(3) 0.03 (4) 0.06

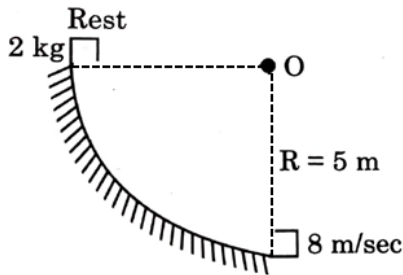
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PHASE - MEB

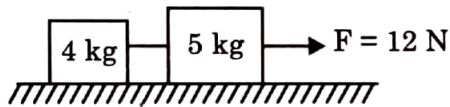
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10. A particle of mass 2 kg is released as shown on circular track of radius $R = 5$ m, velocity of particle at bottom is 8 m/sec then work against friction will be :

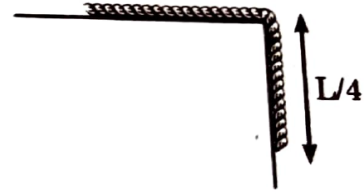


- (1) 64 J (2) 100 J
(3) 36 J (4) 18 J
11. In the given arrangement two blocks are placed on a horizontal surface such that coefficient of friction between the blocks and the floor is $\mu = 0.2$, then tension in the thread connecting the two blocks is : ($g = 10 \text{ m/s}^2$)

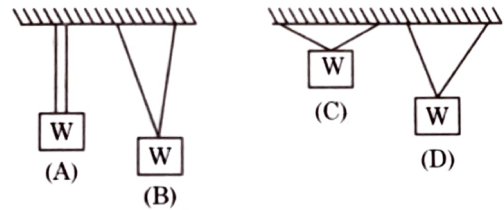


- (1) Zero (2) 2 N
(3) 4 N (4) 6 N
12. $\vec{F} = (2x\hat{i} + 3y\hat{j})$ is working on a particle which moves from (0, 0) to (2, 0) along x-axis and (2, 0) to (2, 2) along y-axis. Find the work done by the force.
- (1) 6 J (2) 4 J
(3) 10 J (4) None
13. The potential energy of an object is given by $U(x) = 3x^2 - 2x^3$, where U is in joules and x is in meters.
- (1) $x = 0$ is stable and $x = 1$ is unstable.
(2) $x = 0$ is unstable and $x = 1$ is stable.
(3) $x = 0$ is stable and $x = 1$ is stable.
(4) $x = 0$ is unstable and $x = 1$ is unstable.

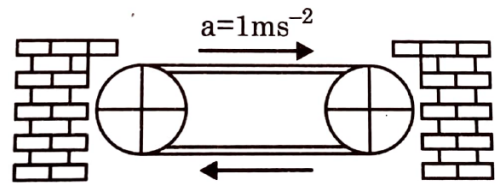
14. A rope of mass M and length L is hanging on table with $1/4^{\text{th}}$ length as shown, then work required to pull it on table will be :



- (1) $\frac{MgL}{4}$ (2) $\frac{MgL}{32}$
(3) $\frac{MgL}{16}$ (4) $\frac{MgL}{8}$
15. Two particles of mass 1 kg and 4 kg have same momentum, calculate ratio of their K.E.
- (1) 4 : 1 (2) 16 : 1 (3) 1 : 4 (4) 8 : 1
16. A weight can be hung in any of the following four ways by string of same type. In which case is the string most likely to break ?

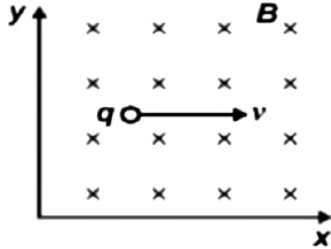


- (1) A (2) B (3) C (4) D
17. A man of mass 60 kg is standing on a horizontal conveyor belt. When the belt is given an acceleration of 1 ms^{-2} , the man remains stationary with respect to the moving belt. If $g = 10 \text{ ms}^{-2}$, the net force on man exerted by belt.



- (1) Zero (2) 603 N
(3) 60 N (4) 600 N
18. The resistance $R = \frac{V}{I}$, where $V = (800 \pm 8) \text{ Volt}$ and $I = (100 \pm 3) \text{ ampere}$. Then total percentage error in resistance is :-
- (1) 7 % (2) 4 % (3) 2 % (4) 8 %

26. A positive charge particle with charge q is moving with speed v in a region of uniform magnetic field B at the instant shown in figure. An external electric field is to be applied so that the charged particle follows a straight line path. The magnitude and direction of electric field required are respectively

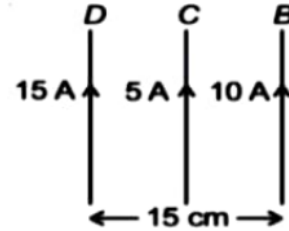


- (1) qvB , $+y$ axis (2) qvB , $-y$ axis
 (3) vB , $-y$ axis (4) $\frac{vB}{q}$, $-x$ axis
27. If a proton enters perpendicularly a magnetic field with velocity v , then time period of revolution is T . If proton enters with velocity $2v$, then time period will be
- (1) T (2) $2T$ (3) $3T$ (4) $4T$
28. A wire is bent in the form of an equilateral triangle of side 100cm and carries a current of 2A . It is placed in a magnetic field of induction 2.0T directed perpendicular into the plane of paper. The direction and magnitude of magnetic force acting on each side of the triangle will be

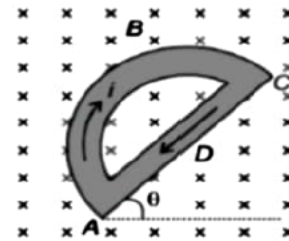


- (1) 2 N , normal to the side towards centre of the triangle
 (2) 2 N , normal to the side away from the centre of the triangle
 (3) 4 N , normal to the side towards centre of the triangle
 (4) 4 N , normal to the side away from the centre of the triangle

29. Three long, straight and parallel wires carrying currents are arranged as shown in the figure. The wire C which carries a current of 5A is so placed that it experiences no force. The distance of wire C from wire D is then

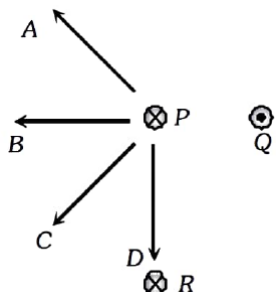


- (1) 9 cm (2) 7 cm (3) 5 cm (4) 3 cm
30. A circular wire ABC and a straight conductor ADC are carrying current i and are kept in the magnetic field B then considering points A and C



- (1) Magnitude of force on ABC is more than ADC
 (2) Magnitude of force on ABC is less than ADC
 (3) Magnitude of force on ABC is equal to that on ADC
 (4) All of these
31. A magnet when placed perpendicular to a uniform field of strength 10^{-4} Wb/m^2 experiences a maximum couple of moment $4 \times 10^{-5}\text{ N/m}$. What is its magnetic moment?
- (1) $0.4\text{ A} \times \text{m}^2$ (2) $0.2\text{ A} \times \text{m}^2$
 (3) $0.04\text{ A} \times \text{m}^2$ (4) $0.06\text{ A} \times \text{m}^2$
32. Points A and B are situated perpendicular to the axis of a 2cm long bar magnet at large distances x and $3x$ from its centre on opposite sides. The ratio of the magnetic fields at A and B will be approximately equal to
- (1) $1 : 9$ (2) $2 : 9$ (3) $27 : 1$ (4) $9 : 1$

33. In figure shows three long straight wires P, Q and R carrying currents normal to the plane of the paper. All three currents have the same magnitude. Which arrow best shows the direction of the resultant force on the wire P



- (1) A (2) B (3) C (4) D
34. An electron moves with speed 2×10^5 m/s along the positive x-direction in the presence of a magnetic induction $B = \hat{i} + 4\hat{j} - 3\hat{k}$ (in Tesla.) The magnitude of the force experienced by the electron in Newton's is (charge on the electron = 1.6×10^{-19} C)
- (1) 1.18×10^{-13} (2) 1.28×10^{-13}
 (3) 1.6×10^{-13} (4) 1.72×10^{-13}
35. A current of 5 ampere is flowing in a wire of length 1.5 metres. A force of 7.5 N acts on it when it is placed in a uniform magnetic field of 2 Tesla. The angle between the magnetic field and the direction of the current is
- (1) 30° (2) 45° (3) 60° (4) 90°

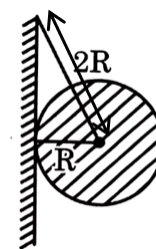
SECTION-B (PHYSICS)

36. What will be the angular width of central maxima in Fraunhofer diffraction when light of wavelength $6000 \text{ \AA}$ is used and slit width is $12 \times 10^{-5} \text{ cm}$.
- (1) 2 rad (2) 3 rad
 (3) 1 rad (4) 8 rad
37. Polarisation of light proves the :
- (1) corpuscular nature of light
 (2) quantum nature of light
 (3) transverse wave nature of light
 (4) longitudinal wave nature of light

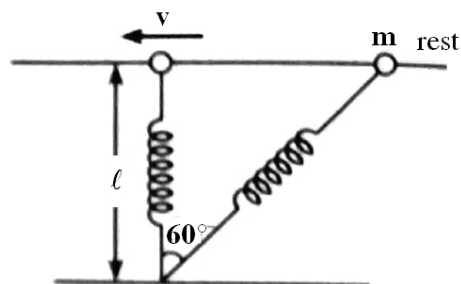
38. Two monochromatic waves each of intensity I have a constant phase difference of ϕ . If these waves superpose, then the intensity of the resultant wave is :

- (1) 4I (2) $4I \cos \phi$
 (3) $4I \cos^2 \phi$ (4) $4I \cos^2 (\phi/2)$

39. A uniform disk of radius R and mass m is connected to a wall by string of length 2R. The normal reaction of wall is :

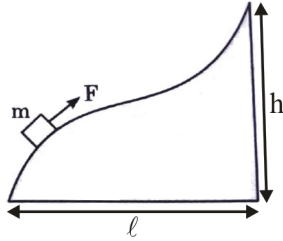


- (1) mg (2) $\frac{mg}{2}$ (3) $\frac{mg}{\sqrt{3}}$ (4) 2 mg
40. A bead of mass m is attached at the end of spring of spring constant k. Its natural length is ℓ . Now spring is stretched on a horizontal rod until its angle with vertical line becomes 60° , then it released velocity of bead in vertical position of string will be :

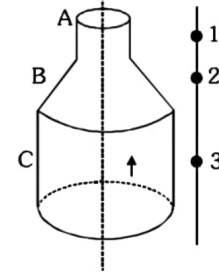


- (1) $\sqrt{\frac{3k\ell^2}{m}}$ (2) $\sqrt{\frac{k\ell^2}{m}}$ (3) $\sqrt{\frac{k\ell^2}{2m}}$ (4) $\sqrt{\frac{3k\ell^2}{2m}}$
41. How much water a pump of 2 kW can raise in one minute to a height of 10 m, take $g = 10 \text{ m/s}^2$?
- (1) 1000 litres (2) 1200 litres
 (3) 100 litres (4) 2000 litres
42. Two springs of spring constants 1500 N/m and 3000 N/m respectively are stretched with the same force. They will have potential energy in the ratio:
- (1) 4 : 1 (2) 1 : 4 (3) 2 : 1 (4) 1 : 2

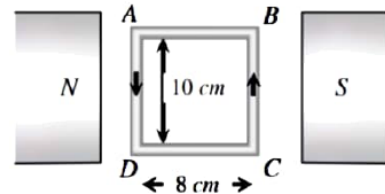
43. A body of mass m is slowly pulled up the hill by a force F which at each point was directed along the tangent of the trajectory as shown in figure. Surface of incline plane is rough (μ). Find the work performed by this force.



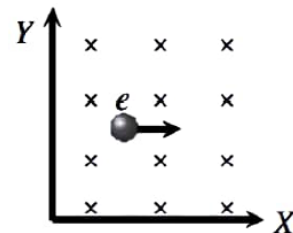
- (1) $mg\ell$ (2) mgh
 (3) $mgh + \mu mg\ell$ (4) Zero
44. If energy (E), velocity (V) and time (T) are chosen as the fundamental quantities, the dimensional formula of surface tension will be (surface tension = $\frac{F}{\ell}$):
- (1) $[EV^{-1}T^{-2}]$ (2) $[EV^{-2}T^{-2}]$
 (3) $[E^{-2}V^{-1}T^{-3}]$ (4) $[EV^{-2}T^{-1}]$
45. A spherical ball of mass m and radius ' r ' is allowed to fall in a medium of viscosity η . The time in which the velocity of the body increases from zero to 0.63 times the terminal velocity is called time constant (τ). dimensionally τ can be represented by
- (1) $\frac{mr^2}{6\pi\eta}$ (2) $\sqrt{\frac{6\pi mr\eta}{g^2}}$
 (3) $\frac{m}{6\pi\eta r}$ (4) None of these
46. A 50-turn circular coil of radius 2.0 cm carrying a current of 5.0 A is rotated in a magnetic field of strength 0.20 T. In a particular position of the coil, the torque acting is half of the maximum torque. The angle between the magnetic field and the plane of the coil is :-
- (1) 60° (2) 30° (3) 90° (4) 100°
47. A long, straight, hollow conductor tube carrying a current has two sections A and C of unequal cross section joined by a conical section B. 1, 2 and 3 are the points on a line parallel to the axis of the conductor. The magnetic fields at 1, 2 and 3 have magnitude B_1 , B_2 and B_3 :-



- (1) $B_1 = B_2 = B_3$ (2) $B_1 = B_2 \neq B_3$
 (3) $B_1 < B_2 < B_3$ (4) $B_1 > B_2 > B_3$
48. A particle of mass m carrying charge q is accelerated by a potential difference V . It enters perpendicularly in a region of uniform magnetic field B and executes circular arc of radius R , then $\frac{q}{m}$ equals
- (1) $\frac{2V}{B^2R^2}$ (2) $\frac{V}{2BR}$ (3) $\frac{VB}{2R}$ (4) $\frac{mV}{BR}$
49. A 100 turns coil shown in figure carries a current of 2 amp in a magnetic field $B = 0.2 \text{ Wb/m}^2$. The torque acting on the coil is



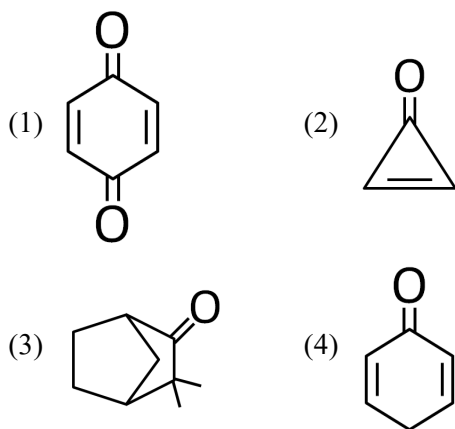
- (1) 0.32 Nm tending to rotate the side AD out of the page
 (2) 0.32 Nm tending to rotate the side AD into the page
 (3) 0.0032 Nm tending to rotate the side AD out of the page
 (4) 0.0032 Nm tending to rotate the side AD into the page
50. In the given figure, the electron enters into the magnetic field. It deflects in direction



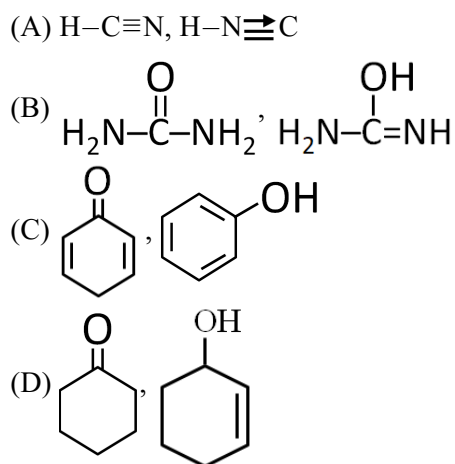
- (1) +ve X direction (2) -ve X direction
 (3) +ve Y direction (4) -ve Y direction

SECTION-A (CHEMISTRY)

51. Which of the following will show tautomerism?

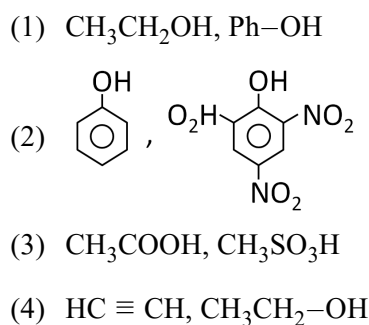


52. Which pair represents tautomer?

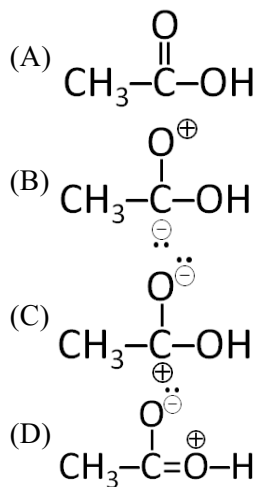


- (1) A,B,C,D
(2) B,D
(3) A,B,C
(4) A,B

53. Which of the following pair can be distinguished using NaHCO_3 ?

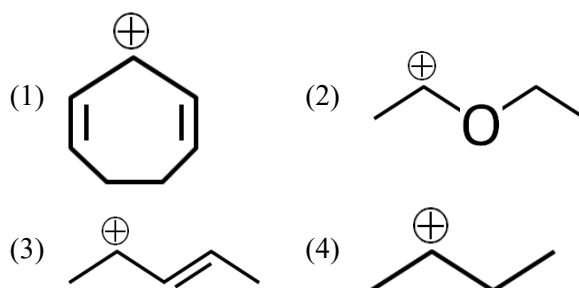


54. Correct stability order of following resonating structures is

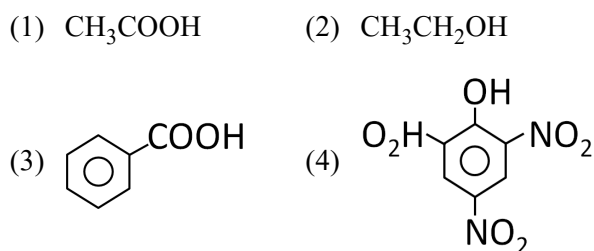


- (1) $\text{A} > \text{B} > \text{C} > \text{D}$ (2) $\text{B} > \text{A} > \text{C} > \text{D}$
(3) $\text{D} > \text{A} > \text{C} > \text{B}$ (4) $\text{A} > \text{D} > \text{C} > \text{B}$

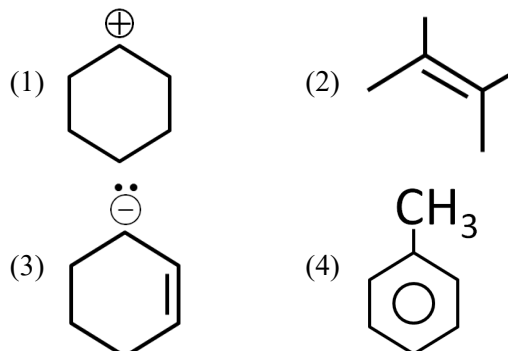
55. Most stable carbocation is ?



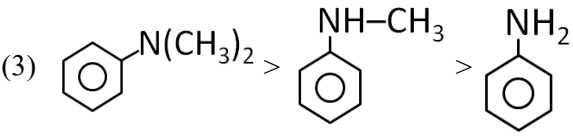
56. Compound with lowest pK_a value is?



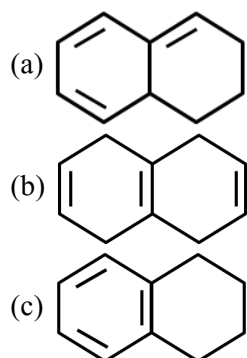
57. Hyperconjugation is not observed in?



58. Correct basic strength comparison?

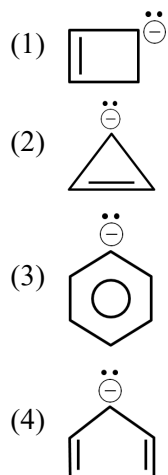
- (1) $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N}$
 (2) $(\text{CH}_3\text{CH}_2)_2\text{NH} > (\text{CH}_3\text{CH}_2)_3\text{N} > \text{CH}_3\text{CH}_2\text{NH}_2$
 (3) 
 (4) All of the above

59. Correct order of heat of hydrogenation for

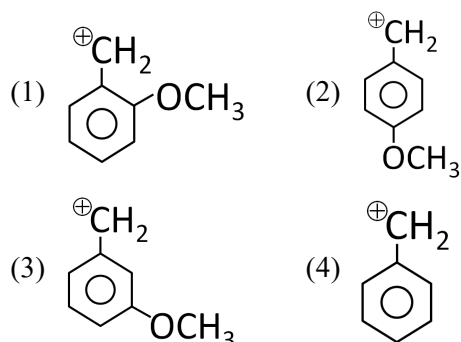


- (1) $a > b > c$ (2) $c > b > a$
 (3) $b > a > c$ (4) $a > c > b$

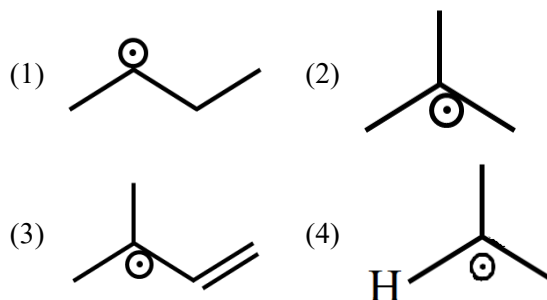
60. Most stable carboanion is?



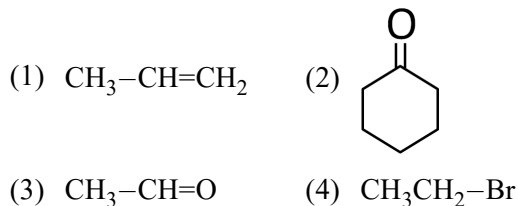
61. Most stable carbocation is :



62. Most stable carbon free radical is :



63. Electrometric effect is not observed in?



64. The molal elevation constant of water = $0.52 \text{ K kg mol}^{-1}$.
 The boiling point of 1.0 molal aqueous KCl solution (assuming complete dissociation of KCl), should be :-

- (1) 100.52°C (2) 101.04°C
 (3) 99.48°C (4) 98.96°C

65. For equal molal concentration of glucose, NaCl, BaCl_2 and AlCl_3 the correct order of boiling point is :-

(Consider complete dissociation of all electrolyte)

- (1) Glucose $>$ $\text{AlCl}_3 >$ $\text{BaCl}_2 = \text{NaCl}$
 (2) $\text{AlCl}_3 >$ $\text{BaCl}_2 >$ $\text{NaCl} >$ glucose
 (3) $\text{BaCl}_2 = \text{NaCl} = \text{glucose} = \text{AlCl}_3$
 (4) $\text{NaCl} >$ $\text{BaCl}_2 >$ $\text{AlCl}_3 >$ glucose

66. What is the molarity of "11.2 V" H_2O_2 solution ?

- (1) 1 M (2) 2 M
 (3) 5.6 M (4) 11.2 M

67. Which of the following is incorrect for ideal solution?

- (1) $\Delta P = P_{\text{obs}} - P_{\text{calculated}} = 0$
 (2) $\Delta G_{\text{mix}} = 0$
 (3) $\Delta H_{\text{mix}} = 0$
 (4) $\Delta U_{\text{mix}} = 0$

68. At 293 K, vapour pressure of pure benzene is 75 mm of Hg and that of pure toluene is 22 mm of Hg. The vapour pressure of the solution which contains 20 mol% benzene and 80 mol % toluene is

- (1) 32.6 mm Hg (2) 64.4 mm Hg
(3) 97 mm Hg (4) 3.26 mm Hg

69. On the basis of information given below mark the correct option.

- (I) In bromoethane and chloroethane mixture, intermolecular interactions of A-A and B-B type are nearly same as A-B type interactions.
(II) In ethanol and acetone mixture A-A or B-B type intermolecular interactions are stronger than A-B type interactions.
(III) In chloroform and acetone mixture A-A or B-B type intermolecular interactions are weaker than A-B type interactions.

- (1) Solution II and III will follow Raoult's law
(2) Solution I will follow Raoult's law
(3) Solution II will show negative deviation from Raoult's law
(4) Solution III will show positive deviation from Raoult's law

70. For an ideal solution with $P_A^0 > P_B^0$, which of the following is true?

- (1) $(\chi_A)_{\text{liquid}} < (\chi_A)_{\text{vapour}}$
(2) $(\chi_A)_{\text{liquid}} > (\chi_A)_{\text{vapour}}$
(3) $(\chi_A)_{\text{liquid}} = (\chi_A)_{\text{vapour}}$
(4) $(\chi_A)_{\text{liquid}}$ and $(\chi_A)_{\text{vapour}}$ cannot be correlated with each other

71. **Assertion** : When NaCl is added to water a depression in freezing point is observed.

Reason : The lowering in vapour pressure causes depression in the freezing point.

- (1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
(2) **Assertion** is true but **Reason** is false.
(3) **Assertion** is false but **Reason** is true.
(4) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

72. According to Henry's law, the partial pressure of gas (P_g) is directly proportional to mole fraction of gas in liquid solution, $P_{\text{gas}} = K_H \cdot X_{\text{gas}}$, where K_H is Henry's constant. Which is incorrect ?

- (1) K_H is characteristic constant for a given gas solvent system
(2) Higher is the value of K_H , lower is solubility of gas for a given partial pressure of gas
(3) K_H has temperature dependence
(4) K_H decreases with increase of temperature.

73. Which of the following mixtures of solutions can function as a buffer solution ?

- (1) 50 ml of 0.2 M NH_3 + 50 ml of 0.1 M HCl
(2) 50 ml of 0.2 M NH_3 + 50 ml of 0.2 M HCl
(3) 50 ml of 0.2 M HCl + 50 ml of 0.2 M NaOH
(4) 50 ml of 0.1 CH_3COOH + 50 ml of 0.1 M HCl

74. Which indicator is suitable for titration of CH_3COOH and NaOH :

- (1) Phenolphthalein (2) Litmus
(3) Methyl orange (4) All

75. On mixing 50 ml of 0.04 M HCl and 50 ml of 0.02 M NaOH, pH of resulting solution would be:-

- (1) 1 (2) 2
(3) 1.7 (4) None

76. Which salt gives highest pH in water :

- (1) Na_2CO_3 (2) NaCl
(3) CuSO_4 (4) KCl

77. A monoprotic acid in 1.00 M solution is 0.001% ionised. The dissociation constant of acid is :

- (1) 1.0×10^{-3} (2) 1.0×10^{-6}
(3) 1.0×10^{-8} (4) 1.0×10^{-10}

78. pH of 10^{-8} M NaOH is nearly equal to :

- (1) 8 (2) 7.04 (3) 6.9 (4) 6

79. How many of the following salts when dissolved in H_2O show neutral behaviour?
 KCl , $NaNO_3$, CH_3COONa , K_2SO_4 , $NaCN$, NH_4Cl .

- (1) 2 (2) 3 (3) 4 (4) 5

80. Which salt doesn't hydrolysed :

- (1) $NaCN$
 (2) $NaCl$
 (3) Na_2S
 (4) NH_4Cl

81. **Assertion (A)** :- Aqueous solution of NH_4Cl is acidic.

Reason (R) :- NH_4^+ ions react with water to produce H_3O^+ .

- (1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
 (2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
 (3) **Assertion** is true but **Reason** is false.
 (4) **Assertion** is false but **Reason** is true.

82. For which of the following reaction, increase in pressure favours the forward reaction?

- (1) $Ni(CO)_4(g) \rightleftharpoons Ni(s) + 4CO(g)$
 (2) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$
 (3) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
 (4) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

83. Conjugate base of HCO_3^- is :-

- (1) H_2CO_3 (2) CO_2 (3) CO_3^{2-} (4) CO

84. Which of the given solution has $pOH = pK_b$

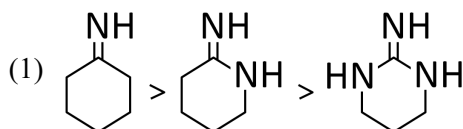
- (1) 10mL, 0.1M HCl + 10mL, 0.2M NH_3
 (2) 20mL, 0.1M HCl + 10mL, 0.1M NH_3
 (3) 50mL, 0.1M HCl + 25mL, 0.1M NH_3
 (4) 25mL, 0.2M HCl + 50mL, 0.1M NH_3

85. K_{a1} and K_{a2} of H_2CO_3 are 10^{-5} and 10^{-9} respectively. pH of 0.1M H_2CO_3 is :-

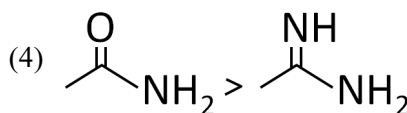
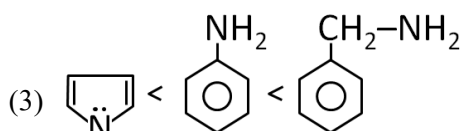
- (1) 7 (2) 3 (3) 5 (4) 11

SECTION-B (CHEMISTRY)

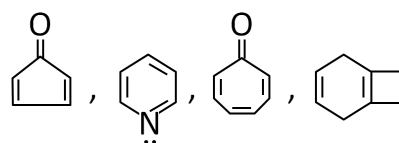
86. Which of the following is correct basic strength order :



- (2) $Et_2NH < Et_3N < EtNH_2$

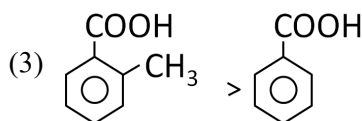
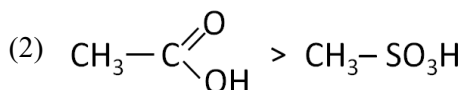
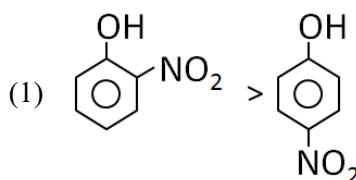


87. How many of given are aromatic?



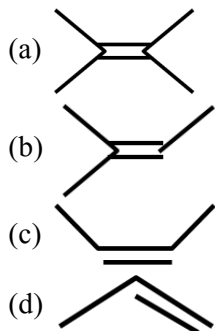
- (1) 3 (2) 4
 (3) 5 (4) 6

88. Correct acidic strength comparison?



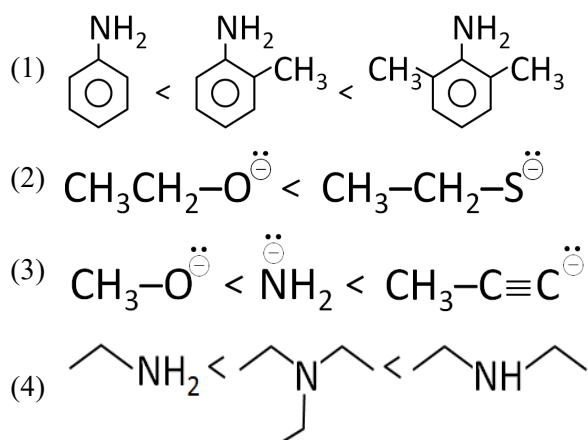
- (4) $CHF_3 > CHCl_3$

89. Correct order of heat of combustion



- (1) $a > b > c > d$ (2) $d > c > b > a$
 (3) $c > b > a < d$ (4) $a > d > b > c$

90. Correct basic strength order is



91. What should be the density of an aqueous solution of urea (molar mass = 60 g/mol) such that the molality and molarity of the solution become equal to 1?

- (1) 1.0 g/ml (2) 1.6 g/ml
 (3) 1.06 g/ml (4) 1.16 g/ml

92. The osmotic pressure of a solution at 0°C is 2 atm. What will be its osmotic pressure at 273°C under similar conditions?

- (1) 0.5 atm (2) 2×237 atm
 (3) 4 atm (4) $273/2$ atm

93. Which solution will have the highest boiling point :-

- (1) 1% glucose in water
 (2) 1% sucrose in water
 (3) 1% NaCl in water
 (4) 1% CaCl_2 in water

94. The vapour pressures of pure A and B at given temperature are 108 and 36 bar respectively what is the mole fraction of B in the vapour phase of the solution containing equimolar mixture of A and B -

- (1) 0.25 (2) 0.5 (3) 0.75 (4) 0.66

95. Calculate the pH of a buffer prepared by mixing 300 cc of 0.3 M NH_3 and 500 cc of 0.5 M NH_4Cl . K_b for $\text{NH}_3 = 1.8 \times 10^{-5}$

- (1) 5.18 (2) 9.82 (3) 8.82 (4) 6.82

96. 20 ml of 0.2 M NaOH is added to 50 ml, of 0.2 M CH_3COOH to give 70ml, of the solution. What is the pH of the solution ? The ionization constant of acetic acid is 2×10^{-5} :-

- (1) 4.522 (2) 5.568 (3) 6.522 (4) 7.568

97. Degree of hydrolysis of 0.01 M $\text{CH}_3\text{COONH}_4(\text{aq})$ is : ($K_a = 10^{-5}$, $K_b = 10^{-5}$)

- (1) 0.01 (2) 0.1 (3) 10^{-3} (4) 10^{-4}

98. Which of the following is incorrect for dissociation constant for acids (K_a)

- (1) It depends on temperature
 (2) K_a is unitless
 (3) Higher the value of K_a , higher will be strength of acid.
 (4) For strong acids value of K_a will be less than 1

99. In which of the following respective ratio should 0.1 M NH_4OH solution and 0.1M HCl solution be mixed, so that the resulting solution behaves like a buffer solution

- (1) 1 : 1 (2) 2 : 1
 (3) 1 : 2 (4) 1 : 3

100. pH of 0.2M aqueous solution of sodium butyrate is :- (Given : K_a of butyric acid = 2×10^{-5})

- (1) 5 (2) 6 (3) 9 (4) 10

Topic : BIOTECHNOLOGY APPLICATION, MICROBES IN HUMAN WELFARE, PLANT BREEDING, BIOMOLECULES, BODY FLUID AND CIRCULATION, BREATHING AND EXCHANGE OF GASES, CELL WALL, CELL MEMBRANE AND ENDOPLASMIC RETICULUM [BACK UNIT] : ORIGIN EVOLUTION AND DOA

SECTION-A (BIOLOGY-I)

101. In MOET:

- (1) Genetic cow is administered hormones, with progesterone-like activity
- (2) The embryos are transferred at 8-32 cells stages into surrogate mothers
- (3) This is the best method for improving the production of animals which are sterile
- (4) Genetic cow releases one ovum per cycle in this technique

102. Leghorn is an exotic breed of:

- (1) Sheep
- (2) Cattle
- (3) Hen
- (4) Buffalo

103. Which is an example of cross-breeding ?

- (1) Mating between donkey and horse
- (2) Mating between bull and cow of same breed
- (3) Mating between Bikaneri ewe and Marino ram
- (4) Mating between superior animals of same breed

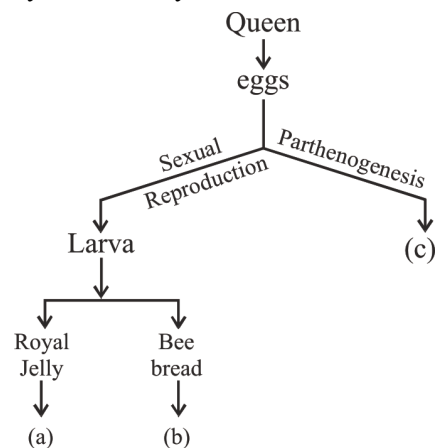
104. Read the following statements and answer the given below questions :

- (a) Mating of more closely related individuals within same breed for 4-6 generations
- (b) increases homozygosity
- (c) Exposes harmful recessive genes
- (d) accumulation of superior genes
- (e) Hissardale
- (f) Desirable qualities of two different breeds
- (g) reduces fertility and productivity
- (h) Mating within the same breed, but having no common ancestor
- (i) Best method for animals that are below average milk productivity
- (j) Herd improvement

How many of the above statements applicable to outcrossing ?

- (1) 4
- (2) 5
- (3) 2
- (4) 1

105. Identify the organism (a), (b) and (c) in the life cycle of honey bee shown below:-



- (1) a = Queen, b = Drone, c = worker
- (2) a = Drone, b = worker, c = Queen
- (3) a = worker, b = Drone, c = Queen
- (4) a = Queen, b = worker, c = Drone

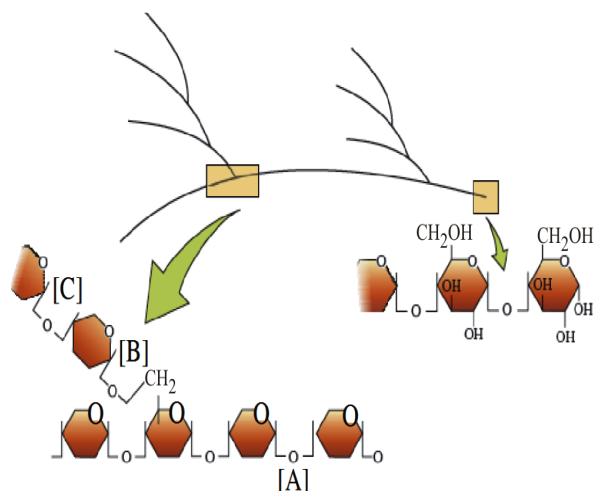
106. _____ refers to the mating of more closely related individuals within same breed for 4-6 generation.

- (1) Out breeding
- (2) In breeding
- (3) Out cross
- (4) Cross breeding

107. Select the incorrect option.

- (1) Wax is primary product of apiculture.
- (2) Bee keeping is labour intensive.
- (3) Apis indica is common species.
- (4) Female honey bees are not example of parthenogenesis.

108.



In the above diagrammatic representation of a portion of glycogen, glycosidic bonds represented by A, B and C are :-

	A	B	C
(1)	α -1'-4''	α -1'-4''	β -1'-4''
(2)	β -1'-6''	β -1'-4''	β -1'-4''
(3)	α -1'-6''	α -1'-4''	α -1'-4''
(4)	α -1'-4''	α -1'-6''	α -1'-4''

109. Which of the following pair of disaccharide is reducing sugar ?

- (1) Sucrose, Maltose (2) Maltose, Trehalose
(3) Sucrose, Trehalose (4) Maltose, Lactose

110. Most abundant type of configuration of protein is :-

- (1) Primary (2) Secondary
(3) Tertiary (4) Quaternary

111. Match the following :-

	Column-I		Column-II
a	Alkaloid	I	Vinblastin, curcumin
b	Essential oils	II	Morphine, Codeine
c	Toxins	III	Lemon grass oil
d	Drugs	IV	Abrin, Ricin

- (1) a-II, b-III, c-IV, d-I
(2) a-III, b-II, c-IV, d-I
(3) a-II, b-III, c-I, d-IV
(4) a-III, b-II, c-I, d-IV

TG: @Chalnaayaaar

112. 'Sonalika' is the improved variety of which crop :-

- (1) Maize (2) Pea
(3) Rice (4) Wheat

113. Which variety of bhindi is resistant to yellow mosaic virus :-

- (1) Pusa sawani (2) Parbhani kranti
(3) Pusa A-4 (4) Pusa komal

114. Which bacterium is used for bioremediation ?

- (1) *Agrobacterium*
(2) *Bacillus thuringiensis*
(3) *Pseudomonas putida*
(4) *E.coli*

115. Which of the following statement is incorrect :-

- (1) Lactic acid bacteria play beneficial role in cheking disease causing microbes in stomach.
(2) Roquefort cheese are ripened by growing a specific bacteria on them
(3) large holes in swiss cheese are due to production of large amount of CO₂.
(4) *Saccharomyces cerevisiae* is known as baker's yeast.

116. A bioactive molecule which is used as immunosuppressive agent in organ-transplant patient is :

- (1) Penicillin (2) Cyclosporin-A
(3) Concavalin-A (4) Streptokinase

117. Which of the following is incorrect regarding baculoviruses

- (1) Species specific
(2) Are from genus-Nucleopolyhedrovirus.
(3) Broad spectrum insecticidal
(4) Helps in integrated pest management programme

118. Which of the following is not Biocontrol Agent ?
- (1) Ladybird
 - (2) *Bacillus thuringiensis*
 - (3) *Trichoderma*
 - (4) *Propionibacterium Sharmanii*
119. **Statement-I** :- Statins act by competitively inhibiting the enzyme responsible for synthesis of cholesterol.
Statement-II :- Statins are commercially used as blood cholesterol lowering agent.
- (1) Statement I and II both are correct.
 - (2) Statement I and II both are incorrect.
 - (3) Statement-I is correct & Statement-II is incorrect.
 - (4) Statement-I is incorrect & Statement-II is correct.
120. Which of the following statement is incorrect?
- (1) transgenic mice are used to test safety of polio vaccine
 - (2) Transgenic animal are use to study toxicity of drugs
 - (3) First transgenic Cow, Rosie produces human protein α -1 antitrypsin.
 - (4) Transgenic animals are used to study gene regulation and their effect on normal body function.
121. Which of the following is natural genetic engineer?
- (1) *Bacillus subtilis*
 - (2) *Escherichia coli*
 - (3) *Pseudomonas putida*
 - (4) *Agrobacterium tumefaciens*
122. Which of the following pair is matched incorrectly.
- (1) Golden rice – Vit. A enriched rice
 - (2) Flavr savr tomato – Pest resistant crop.
 - (3) Bt-cotton – Contain toxic insecticidal protein.
 - (4) RNA interference – *Meloidegyn incognita*

123. Which of the following bacterial specie has high rate of biomass production and growth and can produce 25 tonnes of proteins?
- (1) *Methanobacterium*
 - (2) *Methylophilus methylotrophus*
 - (3) *Meloidegyn incognita*
 - (4) *Mycobacterium avium*
124. (A) Ritual burial
 (B) Developed speech and language centres
 (C) Cave paintings
 (D) Domestication
 (E) Highest cranial capacity
- How many of the above statements are correct for *Homo sapiens* (Cro-magnon man) :-
- (1) 5
 - (2) 4
 - (3) 3
 - (4) 1
125. Match the column and choose the correct option?

	I		II		III
A	<i>Homo habilis</i>	i	1400 CC	a	First man to use fire
B	<i>Homo erectus</i>	ii	650-800 CC	b	Cave painting
C	Neanderthal man	iii	1650 CC	c	First tool maker man
D	Cromagnon man	iv	900 CC	d	Buried their dead

- (1) A-iv-a B-ii-c C-i-b D-iii-d
- (2) A-ii-c B-i-a C-iii-d D-iv-b
- (3) A-ii-c B-iv-a C-i-d D-iii-b
- (4) A-iv-b B-ii-c C-iii-b D-iii-d

126. Arrange the following ancestor of human in chronological order (most primitive first) :-
 (i) Dryopithecus → Australopithecus → Kenyapithecus → Homo erectus
 (ii) Homo habilis → Homo erectus → Cromagnon → Neanderthals.
 (iii) Ramapithecus → Shivapithecus → Neanderthals → Homo erectus.
 (iv) Heidleberg man → Java man → Peking man → Neanderthal man.

- (1) Only iii is correct (2) All are incorrect
 (3) i & iv are correct (4) ii & iii are correct

127. Highest cranial capacity is present in which of following?

- (1) Homohabilis
 (2) Neanderthal
 (3) Cromagnon
 (4) Homoerectus

128. Which of the following is closest to humans:-

- (1) Gibbon
 (2) Gorilla
 (3) *Homo erectus*
 (4) *Dryopithecus*

129. Which one of the following was the first to stand erect & show bipedal movement ?

- (1) Peking man
 (2) Java man
 (3) *Australopithecus*
 (4) Cro-magnon man

130. Which of the following had smallest brain capacity?

- (1) Modern man
 (2) Java man
 (3) Neanderthal man
 (4) *Homo habilis*

131. "Marsupial mole" and Kangaroo are the example of:-

- (1) Divergent evolution
 (2) Convergent evolution
 (3) Adaptive radiation
 (4) Both (1) and (3)

132. Which statement is incorrect about the changes that have occurred on earth since its origin?

- (1) Most of the birds evolved into mammals
 (2) Multicellular organisms evolved from unicellular organisms
 (3) The first organisms that invaded land were plants
 (4) Life originated in the water

133. (A) Ear pinna is vestigial organ in human
 (B) Wings of bats & birds are analogous organs
 (C) Plica circularis & wisdom teeth both are vestigial organ
 (D) Intermediate horse was ruminating.
 Out of the four statements given above :-

- (1) All are incorrect
 (2) Only one is correct
 (3) Only two are incorrect
 (4) Only one is incorrect

134. Select the incorrect statements :-

- (A) The essence of Darwinian theory of evolution is natural selection
 (B) Evolution is a directed process in the sense of determinism
 (C) The geological history of earth is not related with the biological history of earth
 (D) During evolution the rate of appearance of new forms is linked to the life cycle

- (1) A and B (2) B and C
 (3) A and D (4) B and D

135. Following are the two statements regarding the evolution:

- (a) The evolution of life forms has driven by use and disuse of organs.
- (b) Branching descent and natural selection are the two key concepts of Darwinian Theory.

Which one of the following option is correct regarding the above statements?

- (1) Statement (a) is true but (b) is false
- (2) Statement (b) is true but (a) is false
- (3) Both statements (a) and (b) are true
- (4) Both statements (a) and (b) are false

SECTION-B (BIOLOGY-I)

136. Mule is a product of

- (1) Breeding
- (2) Mutation
- (3) Recombination
- (4) Interspecific hybridization

137. How many statement are correct, when cross breeding is a mating between _____ :-

- (A) Two varieties of a breed
- (B) Two varieties of species
- (C) Between two different breeds
- (D) Unrelated members of same species

- (1) 4
- (2) 3
- (3) 2
- (4) 1

138. What is the solution of inbreeding depression?

- (1) Out cross
- (2) In breeding
- (3) MOET
- (4) Cross breeding

139. Arachidonic acid has _____ carbon atoms including the carboxyl carbon atom and has _____ double bonds :-

- (1) 20, 4 (2) 19, 4 (3) 18, 2 (4) 20, 3

140. Resistance to jassids in cotton and cereal leaf beetles in wheat is due to which of the following morphological characteristic ?

- (1) Solid stem
- (2) Hairy leaves
- (3) Nectarless nature
- (4) Both (1) and (2)

141. **Assertion (A)** :- RNAi involves silencing of a specific mRNA due to complementary RNA that binds to & prevents translation of the mRNA.

Reason (R) :- RNAi takes place in all prokaryotic organisms as a method of cellular defense.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

142. Which of the following microbes make plant tolerant to salinity and drought ?

- (1) Anabena
- (2) Nostoc
- (3) Glomus
- (4) Oscillatoria

143. Cry I Ab and Cry II Ab genes proteins are used to control.

- (1) Cotton bollworm, tobacco bollworm
- (2) Corn borer, Armyworm
- (3) Beetles, cornborer
- (4) Corn borer, Cotton bollworms

144. **Statement-I** :- At present 30 recombinant therapeutic have been marketed for human use in India.

Statement-II :- Recombinant therapeutic do not induce unwanted immunological response as is common in case of similar product isolated from non human source

Choose the correct answer from option given below :

- (1) Statement I and II both are correct
- (2) Statement I and II both are incorrect
- (3) Statement-I is correct & Statement-II is incorrect.
- (4) Statement-I is incorrect & Statement-II is correct.

145. Ship of Darwin was

- (1) Beagle
- (2) Queen
- (3) Poodle
- (4) Titanic

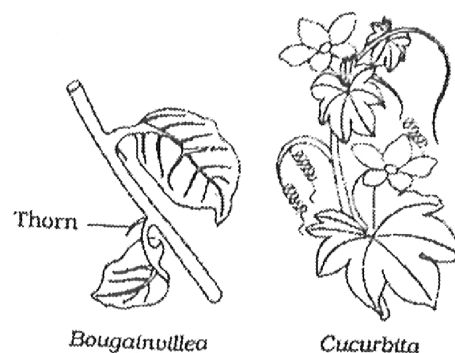
146. What was the most significant trend in the evolution of modern man (*Homo sapiens*) from his ancestors?

- (1) Increasing cranial capacity
- (2) Upright posture
- (3) Shortening of jaws
- (4) Binocular vision

147. The chronological order of human evolution from early to the recent is :-

- (1) *Ramapithecus* → *Homo habilis* → *Australopithecus* → *Homo erectus*
- (2) *Australopithecus* → *Homo habilis* → *Ramapithecus* → *Homo erectus*
- (3) *Australopithecus* → *Ramapithecus* → *Homo habilis* → *Homo erectus*
- (4) *Ramapithecus* → *Australopithecus* → *Homo habilis* → *Homo erectus*

148. Thorn of Bougainvillea and tendrill of Cucurbita represents :



- (1) Convergent evolution
- (2) Analogous organs
- (3) Divergent evolution
- (4) Homoplastic organs

149. When more than one adaptive radiation appeared to have occurred in an isolated geographical area is called:-

- (1) Divergent evolution
- (2) Convergent evolution
- (3) Reproductive isolation
- (4) Saltation

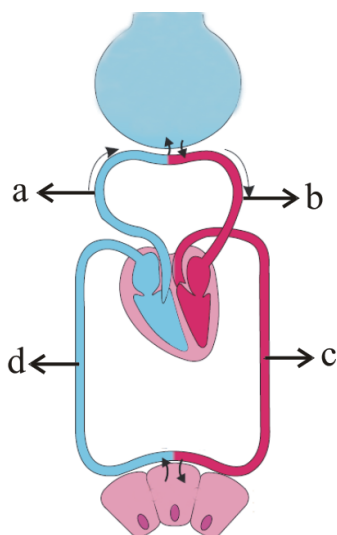
150. The most likely reason for the development of resistant against pesticide in insects damaging the crop is

- (1) Genetic recombination
- (2) Directed mutations
- (3) Acquired heritable germinal changes
- (4) Random mutations

Topic : BIOTECHNOLOGY APPLICATION, MICROBES IN HUMAN WELFARE, PLANT BREEDING, BIOMOLECULES, BODY FLUID AND CIRCULATION, BREATHING AND EXCHANGE OF GASES, CELL WALL, CELL MEMBRANE AND ENDOPLASMIC RETICULUM [BACK UNIT] : ORIGIN EVOLUTION AND DOA

SECTION-A (BIOLOGY-II)

151. Recognise the figure and find out the correct matching.



- (1) a-Systemic arteries, b-Systemic veins, c-Pulmonary artery, d-Pulmonary vein
 - (2) b-Systemic arteries, a-Systemic veins, d-Pulmonary artery, c-Pulmonary vein
 - (3) c-Systemic arteries, d-Systemic veins, a-Pulmonary artery, b-Pulmonary vein
 - (4) d-Systemic arteries, c-Systemic veins, b-Pulmonary artery, a-Pulmonary vein
152. Every 100 ml of deoxygenated blood delivers approximately
- (1) 3 ml of CO_2 to alveoli
 - (2) 4 ml of CO_2 to alveoli
 - (3) 5 ml of CO_2 to alveoli
 - (4) 6 ml of CO_2 to alveoli

153. Pneumotaxic centre can moderate the functions of respiratory rhythm centre by

- (1) Reducing the duration of inspiration
- (2) Increasing the duration of inspiration only
- (3) First increasing and then reducing the duration of expiration
- (4) Increasing the duration of expiration only

154. Which of the following is incorrect match of disorder

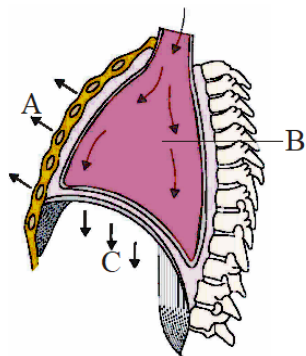
- (1) Occupational respiratory disorders - Asthma
- (2) Angina - Severe pain in chest due to no enough oxygen is reaching the heart muscles
- (3) Cardiac arrest - The heart stops beating
- (4) Renal calculi-Stone or insoluble mass of crystallised salts (oxalates) formed within the kidney

155. Match the following columns and choose the correct answer from the following options :-

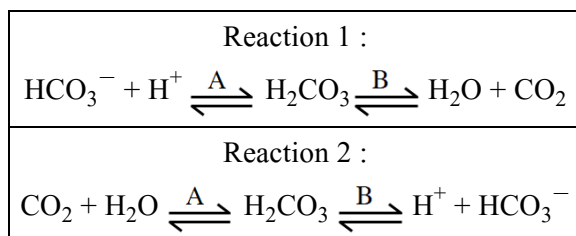
	Volume/capacity		Value in mL
a	Tidal volume	i	2500-3000
b	Residual volume	ii	2100-2300
c	Vital capacity	iii	4000-4500
d	Functional residual capacity	iv	1100-1200
e	Inspiratory reserve volume	v	500

	a	b	c	d	e
(1)	i	ii	iii	iv	v
(2)	v	iv	iii	ii	i
(3)	i	ii	iii	v	iv
(4)	v	iv	iii	i	ii

156. The figure below shows three steps (A, B, C) of mechanism of breathing. Select the option giving correct identification together with what it represents:



- (1) A-Ribs and sternum depressed
B-Volume of lungs increased
C-Diaphragm contracted
- (2) A-Ribs and sternum raised
B-Volume of lungs decreased
C-Diaphragm contracted
- (3) A-Ribs and sternum raised
B-Volume of lungs increased
C-Diaphragm contracted
- (4) A-Ribs and sternum raised
B-Volume of lungs increased
C-Diaphragm relaxed
157. Read the reaction (1 and 2), identify the enzymes A and B and the site of reactions :



Options

	A	B	Site of Reaction-1	Site of Reaction-2
(1)	Carbonate dehydrase	Carbonic anhydrase	Plasma	Alveolar region
(2)	Carbonic anhydrase	Carbonic anhydrase	Alveolar region	RBCs
(3)	Carbonic anydrase	Decarboxylase	Alveolar region	Plasma
(4)	Decarboxylase	Carbonic anhydrase	Plasma	Alveolar region

158. The solubility of CO_2 is _____ times higher than that of O_2 .

- (1) 10-15 (2) 20-25
(3) 30-35 times (4) 210 times

159. About 1200 ml of air left in lungs is called—

- (1) Tidal volume
(2) Inspiratory reserve volume
(3) Residual volume
(4) Vital capacity

160. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R) .

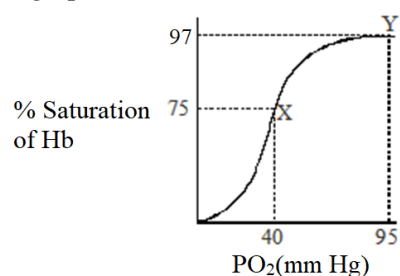
Assertion (A) : Pneumotaxic centre can moderate the functions of Respiratory rhythm centre.

Reason (R) : Neutral signal from this centre can reduce the duration of inspiration.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

161. Oxygen dissociation curve of haemoglobin is shown in graph



- (1) X - Oxygenated blood, Y - Deoxygenated blood
(2) X - De oxygenated blood, Y - Oxygenated blood
(3) X - Blood of Haemophilic person, Y - Blood of normal person
(4) X - Blood of embryo Y - Blood of adult

162. Which of the following statements correctly defines Bohr's effect ?

- (1) Rise in P_{50} with a decrease in CO_2 concentration
- (2) Rise in P_{50} with increase in pH
- (3) Rise in P_{50} with a decrease in pH
- (4) Fall in P_{50} with a decrease in pH

163. Choose the correct statements :

- (1) Pneumotaxic present in pons can increase the duration of inspiration
- (2) Role of O_2 in the regulation of respiration is significant
- (3) High CO_2 concentration in blood causes oxyhaemoglobin dissociation
- (4) 100 ml deoxygenated blood delivers approximately 4ml CO_2 per day

164. When in a human being each ventricle pumps out approximately 50 ml of blood but heart beat is 72 times min^{-1} then value of cardiac output is :-

- (1) 5 L per minute
- (2) 5000 ml per minute
- (3) 3.6 L per minute
- (4) 5.5 L per minute

165. First heart sound lub produced during :-

- (1) Beginning of ventricular systole
- (2) Beginning of ventricular diastole
- (3) Beginning of atrial systole
- (4) Beginning of joint diastole

166. Phagocytic blood cells are :-

- (1) Neutrophil & Acidophil
- (2) Basophil & neutrophil
- (3) Acidophil & Basophil
- (4) Monocyte and neutrophil

167. Semilunar valves are present between :-

- (1) Left ventricle and aorta
- (2) Right ventricle and pulmonary artery
- (3) Left ventricle and aorta and also between right ventricle and pulmonary artery
- (4) Left ventricle and pulmonary aorta and also between right ventricle and aorta

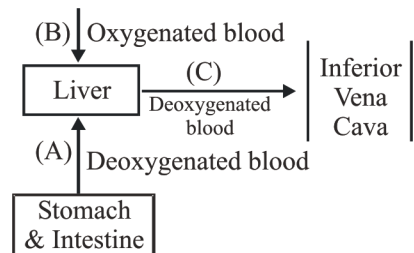
168. Choose the wrong pair :-

	Column-A	Column-B
(1)	RBCs	5 millions – 5.5 millions mm^{-3} of blood
(2)	WBCs	6000 – 8000 mm^{-3} of blood
(3)	Platelets	150000 – 350000 mm^{-3} of blood
(4)	Thrombocytes	6000 – 8000 mm^{-3} of blood

169. Clotting factors are present :-

- (1) In plasma in inactive form
- (2) In serum in inactive form
- (3) In blood in active form
- (4) In plasma in active form

170. Please observe the diagram below



Here vessel A, B and C are –

- (1) A – Hepatic vein (2) A – Portal vein
B – Portal vein B – Hepatic artery
C – Hepatic artery C – Hepatic vein
- (3) A – Hepatic artery (4) A – Portal vein
B – Hepatic vein B – Hepatic vein
C – Portal vein C – Hepatic artery

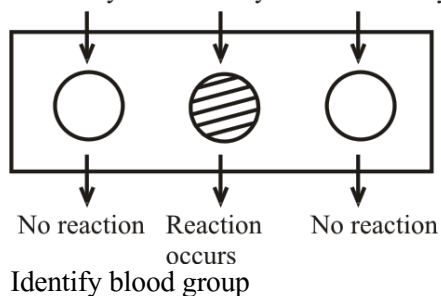
171. Which one of the following option is correctly matched with its category?

	Group	Type of Circulation	Blood is oxygenated by
(1)	Fish	Single circulation	Skin
(2)	Amphibia	Double circulation	Gills
(3)	Reptiles	Incomplete double circulation	Lungs
(4)	Birds	Incomplete double circulation	Lungs

172. Significance of heart valves is

- (1) to allow the flow of blood in one direction , i. e. from ventricles to atria and from the ventricles to the pulmonary artery or aorta
- (2) to allow the flow of blood in one direction , i. e. from atria to ventricles and from the ventricles to the pulmonary artery or aorta
- (3) to allow the flow of blood in one direction , i. e. from atria to ventricles and from the pulmonary artery or aorta to the ventricles
- (4) to allow the flow of blood in one direction , i. e. from ventricles to atria and from the pulmonary artery or aorta to the ventricles

173. There are 3 drops of blood of a human on a slide :-
Antibody a Antibody b Rh Antibody



- (1) B⁺ (2) B⁻ (3) A⁺ (4) A⁻

174. A special neural centre in the _____ can moderate the cardiac function through autonomic nervous system

- (1) S.A. Node
- (2) A.V. Node
- (3) Medulla oblongata
- (4) Both 1 and 2

175. Read the following given statements and choose the correct option :-

Statement-I – Atria receive blood from all parts of the body which subsequently flows to ventricles.

Statement-II – Action potential generated at S.A. node passes to atria and A.V. node.

- (1) Statement-I is wrong and statement-II is correct.
- (2) Statement-II is wrong and statement-I is correct.
- (3) Statement-I and II are correct.
- (4) Statement-I and II are wrong.

176. The systemic circulation provides nutrients, __A__ and other essential substances to the __B__ and takes __C__ and other harmful substances away for elimination. Choose the correct option for A, B and C.

- (1) A-CO₂, B-Tissue, C-O₂
- (2) A-O₂, B-Tissue, C-CO₂
- (3) A-O₂, B-Tissue, C-NO₂
- (4) A-NO₂, B-Tissue, C-CO₂

177. Which of the following structure show multicellularity in plants :-

- (1) Presence of middle lamella
- (2) Presence of plasmodesmata
- (3) Both (1) and (2)
- (4) Presence of cell wall

178. Endoplasmic reticulum, Golgibody, Lysosome, Mitochondria, Nucleus, Ribosome, Chloroplast, Leucoplast, Chromoplast, Vacuole
How many above given cell organelle is single membrane bound?

(1) 4 (2) 3 (3) 5 (4) 2

179. Which of the following statement is not correct :-

- (1) Pentacyclin sterol (Hopanoid) provide the stability to the prokaryotic cell membrane
(2) Plasma membrane have enzymes
(3) Plasma membrane in an asymmetric structure
(4) Tunnel protein may be integral or peripheral protein

180. Which of the following structure is used for invitro protein synthesis study and do not exist as such in the living cell ?

- (1) RER
(2) SER
(3) Ribosome
(4) Microsome

181. Function of Endoplasmic reticulum is/are :-

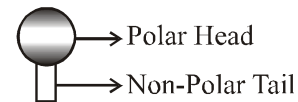
- (a) Detoxification
(b) Formation of Acrosome
(c) Formation of nuclear membrane
(d) Intra cellular exchange of material

- (1) a, b, d
(2) a, c, b
(3) a, c, d
(4) a, b, c, d

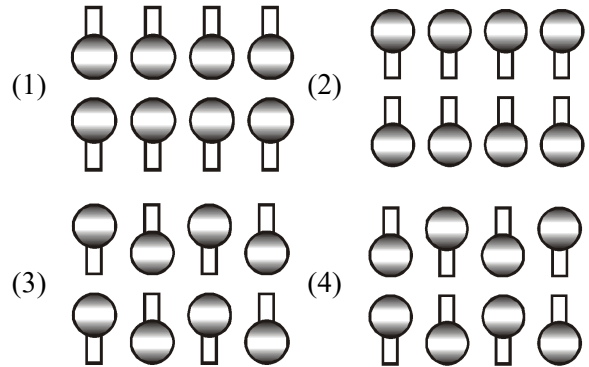
182. The outer membrane of nucleus usually remains continuous with which cell organelle ?

- (1) Mitochondria
(2) Endoplasmic reticulum
(3) Lysosome
(4) Chloroplast

183. The lipid molecules present in plasma membrane have polar heads and non-polar tails :-



Which figure represents the correct arrangement of lipids in lipid bilayer ?



184. Which of the following component is common in all three plant cell wall ?

- (1) Cellulose (2) Pectin
(3) Xylan (4) Hemicellulose

185. Match column I with column II

Column I		Column II	
(A)	Primary wall	(i)	Maintain continuity of cytoplasm
(B)	Secondary wall	(ii)	Common layer between two plant cells
(C)	Middle lamella	(iii)	Absent in meristem cells
(D)	Plasmodesmata	(iv)	Gradually diminishes as cell matures.

- (1) A-ii, B-iv, C-i, D-iii
(2) A-iv, B-iii, C-i, D-ii
(3) A-iii, B-iv, C-ii, D-i
(4) A-iv, B-iii, C-ii, D-i

SECTION-B (BIOLOGY-II)

186. How many statements are correct ?
 (a) First mammals were like shrews.
 (b) In continental drift, South America joined North America.
 (c) Pouched mammals of Australia survived because of lack of competition.
 (d) *Ramapithecus* was more ape like and *Dryopithecus* was more man like.
 (1) 4 (2) 3 (3) 2 (4) 1
187. In relation to the natural selection evolution is the:-
 (1) Process
 (2) Outcome (consequence)
 (3) Mechanism
 (4) Purpose
188. In human respiratory system, conduction zone extends from :-
 (1) External nostril to the terminal bronchioles.
 (2) External nostril to the respiratory bronchioles
 (3) Internal nostril to terminal bronchioles
 (4) Respiratory bronchioles to alveoli
189. Correct sequence of filling of volumes in lungs :-
 (1) $RV \rightarrow ERV \rightarrow IRV \rightarrow TV$
 (2) $ERV \rightarrow RV \rightarrow IC \rightarrow IRV$
 (3) $[TLC - VC] \rightarrow EC \rightarrow IRV$
 (4) $FRC \rightarrow [IC - IRV] \rightarrow ERV$
190. Identify the correct set of statements :
 (a) Partial pressure of CO_2 , $[H^+]$ and temperature are the factors which can interfere with the binding of O_2 with Hb.
 (b) A sigmoid curve is obtained when partial pressure of pCO_2 is plotted against the percentage saturation of haemoglobin with oxygen.

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- (c) Oxygen dissociation curve is highly useful in studying the effect of factors like pCO_2 , $[H^+]$ and temperature on binding of O_2 with haemoglobin.
 (d) In the alveoli, where there is high pCO_2 , low pO_2 , higher $[H^+]$ and low temperature are the factors favourable for the formation of oxy-haemoglobin.
 (e) In tissues low pO_2 , high pCO_2 , high $[H^+]$ and higher temperature conditions are favourable for dissociation of O_2 from Hb. Choose the correct answer from options given below :

- (1) Only a, c and e (2) Only b and d
 (3) a, b and d only (4) a, b, c and d
191. Every 1000 ml of oxygenated blood can deliver around how much ml of O_2 to the tissue under normal physiological conditions?
 (1) 5 mL (2) 1.4 mL (3) 15 mL (4) 50 mL
192. T-wave of ECG represents :-
 (1) Atrial repolarisation
 (2) Ventricular repolarisation
 (3) Atrial depolarisation
 (4) Ventricular depolarisation
193. Heart receives its supply of oxygenated blood via?
 (1) Cardiac portal system
 (2) Pulmonary vein
 (3) Pulmonary artery
 (4) Coronary artery
194. The correct route through which impulse travel in the heart is :-
 (1) S.A. node $\rightarrow$ Purkinje fibres $\rightarrow$ Bundle of His $\rightarrow$ A.V. node $\rightarrow$ Heart muscles
 (2) A.V. Node $\rightarrow$ S.A. Node $\rightarrow$ Purkinje fibres $\rightarrow$ Bundle of His $\rightarrow$ Heart muscles
 (3) S.A. Node $\rightarrow$ A.V. Node $\rightarrow$ Bundle of His $\rightarrow$ Purkinje fibres $\rightarrow$ Heart muscles
 (4) A.V. Node $\rightarrow$ Bundle of His $\rightarrow$ S.A. Node $\rightarrow$ Purkinje fibres

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PHASE - MEB

195. Which of the following events occur during joint diastole?
(a) All four chambers are in relaxed state.
(b) AV valves are open
(c) Semilunar valves are closed
(d) Blood from the pulmonary veins and vena cava flows into the left and right ventricles through the right and left atria respectively.
The correct option.
(1) only 'a' (2) a, b only
(3) a, b, c only (4) a, b, c, d all
196. What are the number of double circulation per min. in human :
(1) 144 (2) 72 (3) 200 (4) 100
197. Integral cell membrane proteins :
(1) are partially embedded in lipid layers
(2) are completely embedded in lipid layers
(3) show lateral but not flip flop movements within bilayer of lipid
(4) all of these
198. "Presence of cell wall is a unique character of the plant cells".
Above mentioned statement was given by -
(1) Schleiden
(2) Schwann
(3) Rudolf virchow
(4) Schleiden and schwann
199. Plasma membrane is :
(1) Permeable
(2) Semipermeable
(3) Selective permeable
(4) Impermeable
200. Which component of plasma membrane are involved in cell to cell recognition mechanism ?
(1) Phospholipid
(2) Cholestrol
(3) Carbohydrates
(4) Integral proteins



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FROM CURRENT & PREVIOUS UNITS

(Session : 2023 - 2024)

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MINOR TEST NO. & DATE	CURRENT UNIT	PREVIOUS UNIT (COMPLETE TOPIC)	
08 10-09-2023	SYLLABUS COVERED FROM 16-08-2023 TO 09-09-2023	P	SHM AND WAVE MOTIONS
		C	ELECTROCHEMISTRY
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